GREEN VALLEY EDUCATIONAL INSTITUTE (GVEI)

DEPARTMENT OF INFORMATION & COMMUNICATION TECHNOLOGY (ICT)

EDUCATOR PROFESSIONAL DEVELOPMENT & COMPETENCY MAPPING (2026–2027)

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1. Purpose and Strategic Vision

The ICT Department recognises that the quality of technology education depends not only on curriculum delivery but also on educators' subject expertise, pedagogical competence, practical proficiency, and commitment to continuous professional growth.

This document establishes a structured framework for reviewing educator competencies, identifying professional strengths, recognising development needs, and translating those needs into measurable improvement actions.

> **CORE DEVELOPMENTAL MANDATE**
> *"The objective is not merely to document qualifications or training attendance. It is to establish a clear connection between educator capability, classroom practice, student engagement, and demonstrated learning outcomes."*

The competency-mapping process is evidence-informed, developmental, and aligned directly with the institutional academic priorities of Green Valley Educational Institute (GVEI).

2. Educator Profile, Expertise and Responsibilities

An individual professional portfolio is maintained for each ICT educator, recording verifiable academic credentials, instructional allocations, and departmental responsibilities.

Information Required	Mandatory Details to Record
Educator Name	Official name and faculty identity code.
Designation	Current institutional designation and wing allocation.
Classes / Levels Assigned	Actual teaching allocation across assigned grades (Levels I–XII).
Subjects / Areas Handled	ICT, Computer Science, Artificial Intelligence, Python coding, or primary IT.
Academic Qualifications	Relevant university degrees, postgraduate diplomas, and accredited credentials.
Teaching Experience	Documented years of professional teaching and laboratory instruction experience.
Professional Training	Completed workshops, technical courses, seminars, and CBSE/industry certifications.
Additional Responsibilities	Laboratory management, club mentoring, examination duties, or CBT coordination.
Areas of Core Expertise	Demonstrated computational proficiencies, programming languages, and

	pedagogical strengths.
Development Priorities	Targeted technical and instructional competencies identified for further enhancement.

Supporting Evidence: Staff profiles, verified degree certificates, training credentials, teaching timetables, and departmental administrative registers.

3. ICT Educator Competency Framework

Faculty proficiencies are evaluated across ten standard domains adapted to each educator's specific grade band and teaching responsibilities:

Competency Domain	Key Institutional Performance Indicators
Subject Knowledge	Accuracy, conceptual depth, curriculum comprehension, and the ability to confidently resolve complex student inquiries.
Practical & Technical Proficiency	Fluent mastery of IDEs, programming languages (Python, SQL, Scratch), operating systems, and laboratory networking.
Pedagogical Competence	Structured lesson design, instructional clarity, adaptive delivery, and orderly classroom/lab management.
Competency-Based Teaching	Designing application-oriented tasks, computational thinking exercises, and meaningful student-led drills.
Assessment Literacy	Formative diagnostic design, rubric authoring, constructive written feedback, and using assessment data to guide reteaching.
Digital & AI Integration	Purposeful, age-appropriate, ethical, and syllabus-aligned integration of digital productivity and AI learning tools.
Student Engagement & Inclusion	Differentiated scaffolding, inclusive participation models, and proactive support for diverse student learning curves.
Professional Practice	Rigorous lesson preparation, punctuality, active peer collaboration, reflective journaling, and departmental duties.
Innovation & Project Mentoring	Capacity to inspire and guide capstone software projects, hackathons, robotics builds, and AI Club initiatives.
Continuous Professional Growth	Receptivity to feedback, active participation in capacity-building workshops, and measurable pedagogical improvement.

> **EVALUATION INTEGRITY DIRECTIVE**
> *Evaluation Benchmark Principle: Competency evaluations are substantiated by concrete classroom evidence, student artifacts, and observation logs rather than arbitrary assumptions, seniority, or academic qualifications alone.*

4. Educator Self-Assessment Protocol

Every educator completes an annual structured self-assessment to reflect on their instructional confidence, technical proficiency, and self-identified professional-development needs.

The standard 4-tier proficiency scale is defined as follows:

- **Level 1 — Developing:** Requires structured supervision, technical mentoring, and foundational pedagogical support.
- **Level 2 — Progressing:** Demonstrates partial competence but requires guided practice and occasional support.
- **Level 3 — Competent:** Autonomously demonstrates expected technical and instructional competencies in routine practice.
- **Level 4 — Advanced:** Exhibits exemplary mastery and actively mentors colleagues in curriculum delivery and innovation.

Faculty Self-Assessment Matrix Template

Competency Area	Self-Rating (1–4)	Documented Evidence / Concrete Example	Specific Development Need
Subject Knowledge	**Level 3 — Competent**	CBSE Class XI/XII curriculum delivery & Python coding logs	Advanced algorithmic data structures (Stacks/Queues)
Practical & Technical Skills	**Level 4 — Advanced**	Setup of MySQL database servers and Python IDE environments	Hands-on containerization and web deployment workflows
Lesson Planning & Delivery	**Level 3 — Competent**	Standardized, approved daily lesson plans & trace-table guides	Refining inquiry-based starter activities and hooks
Competency-Based Teaching	**Level 3 — Competent**	Student-created mini-projects & interactive Scratch simulations	Designing open-ended real-world problem challenges
Assessment & Feedback	**Level 2 — Progressing**	Regular periodic tests and graded lab record books	Rubric-based diagnostic marking and targeted remediation
Digital & AI Integration	**Level 3 — Competent**	Integration of AI literacy modules & Teachable Machine labs	Prompt-engineering methodologies for classroom coding
Student Engagement & Inclusion	**Level 2 — Progressing**	Equal opportunity questioning in laboratory sessions	Differentiated task sheets for slower coding students
Project Mentoring & Innovation	**Level 3 — Competent**	Mentored student teams for inter-school tech symposiums	Structuring multi-week agile sprints for capstone projects
Reflective Practice	**Level 3 — Competent**	Documented post-lesson reflection logs in teacher diaries	Translating reflective insights into revised lesson plans

Developmental Philosophy: The self-assessment represents a constructive baseline, not an audit verdict. Ratings are reviewed collaboratively against classroom observations and pedagogical outcomes.

5. Teaching Strengths and Targeted Improvement Responses

The annual review identifies both instructional strengths to sustain and priority areas requiring focused pedagogical intervention:

Instructional Review Area	Evidence to Consider in Audit	Targeted Departmental Development Response
Conceptual Explanation	Lesson observations, student responses, and coding task success rates.	Subject refresher clinics, code-tracing workshops, and demo-based peer coaching.
Practical Lab Instruction	Laboratory logbooks, terminal error logs, and practical task completions.	Hands-on technical upskilling in modern compilers, IDEs, and network tools.
Lesson Planning Rigor	Learning objective alignment, curriculum	Collaborative departmental planning sessions

	sequencing, and courseware.	and standardized lesson vetting.
Student Engagement & Focus	Active time-on-task, student question distribution, and lab collaboration.	Implementation of pair programming, interactive live polls, and active learning drills.
Assessment Practices	Question paper blueprints, rubric clarity, feedback notes, and mark sheets.	Assessment-design masterclasses, rubric calibration, and marks moderation.
AI & Digital Integration	Documented classroom AI activities and student digital artifacts.	Specialized workshops on safe generative AI integration and computational ethics.
Differentiated Support	Identification of diagnostic gaps and documented remedial interventions.	Training in multi-tiered scaffolding and remedial task sheet development.
Professional Reflection	Reflection diaries and observable subsequent pedagogical adaptations.	Structured monthly reflection reviews and 1-on-1 pedagogical mentoring.

Action Mandate: Every recorded development priority must lead directly to a structured professional-development activity rather than remaining a passive evaluation observation.

6. Classroom Observation and Professional Feedback

Formative classroom observations furnish primary evidence regarding instructional quality and guide subsequent professional-development planning.

Supervisory and peer observation protocols encompass nine evaluative dimensions:

- **Objective Precision:** Clarity of learning objectives and accuracy of technical explanations.
- **Content Accuracy:** Rigorous, error-free demonstration of coding syntax and computing concepts.
- **Pacing & Structure:** Optimal period distribution across explanation, modeling, and lab drills.
- **Demonstration Mastery:** Effective use of projection displays, live coding, and step-by-step deconstruction.
- **Student Practical Time:** Ensuring students spend substantial class time executing tasks independently.
- **Diagnostic Questioning:** Checking comprehension via Socratic questioning and rapid-fire checks.
- **Differentiated Inclusion:** Providing targeted assistance to students experiencing conceptual hurdles.
- **Lab Infrastructure Handling:** Maintaining workstation discipline, network health, and safety protocols.
- **Outcome Attainment:** Gathering verifiable evidence that stated lesson benchmarks were achieved.

> **OBSERVATION RECORDING PROTOCOL**
> *Observation Record Specification: Every formal visit requires documentation of the observation date, assigned class/section, lesson focus, verified strengths, priority growth areas, agreed action items, and a designated follow-up review date. If an observation has not occurred, the record must explicitly state 'Not yet conducted'.*

7. Teaching Demonstrations and Sample Lessons

Teaching demonstrations and peer sample lessons serve as platforms to evaluate instructional clarity, subject command, and curriculum translation in a collaborative setting.

A standard teaching demonstration incorporates a 7-stage sequence:

7.1 Clearly Articulated Learning Objective: Explicitly stating target competencies and curriculum linkage.

7.2 Precise Theoretical Explanation: Deconstructing abstract computing concepts using analogies and correct terminology.

7.3 Live Practical Modeling: Executing a live worked example or debugging demonstration on the main display.

7.4 Guided Student Drill: Leading students through scaffolded coding drills with immediate verbal checks.

7.5 Independent Applied Task: Assigning an unassisted problem-solving challenge or creative design exercise.

7.6 Diagnostic Check for Understanding: Administering a rapid formative assessment (quiz, code check, exit ticket).

7.7 Post-Demonstration Reflection: Conducting a structured self-critique on pedagogical timing and efficacy.

Evidentiary Archive: The department maintains lesson plans, demonstration code repositories, observer evaluation rubrics, and documented peer feedback for all conducted demonstrations.

8. Professional Training and Skills Acquired Register

The department logs all capacity-building workshops, seminars, and technical training courses completed by ICT educators. Entries differentiate strictly between attendance, acquired skills, and verifiable classroom application:

Educator Name	Training / Workshop Title	Provider / Institution	Date	Core Skills Acquired	Classroom Application Evidence
Mehraj U Din Mir	**Computational Thinking & AI in Schools**	GD Goenka / EduPartners	2026-05-18	Algorithmic thinking, CT pedagogy, AI literacy	Introduced unplugged CT puzzle drills in Class VI-VIII
Mehraj U Din Mir	**Advanced Python & Data Science Pedagogy**	IIT Delhi Extension	2026-02-10	Pandas, Matplotlib, Data visualization & SQL	Overhauled Class XII Informatics Practices practical files
Faculty Member A	**CBSE Capacity Building in Computer Science**	CBSE Centre of Excellence	2026-04-12	Latest Board exam marking rubrics & blueprints	Implemented standardized CBSE practical assessment rubrics
Faculty Member B	**Block-based Coding & Creative Computing**	Scratch Foundation	2026-06-22	Interactive game loops, event handling, animations	Integrated interactive game design in Middle School curriculum
Faculty Member C	**Ethical AI & Digital Safety Curriculum**	CyberPeace Foundation	2026-07-15	Cyber etiquette, data privacy, AI ethics	Conducted school-wide Cyber Safety week and poster drive
Faculty Member D	**CBT Software Deployment & Lab Administration**	Institutional IT Cell	2026-03-29	LAN test server setup, secure browser lockdowns	Successfully coordinated pilot batch CBT examinations
Mehraj U Din Mir	**Digital HPC**	SCERT	1st and 2nd OCT	DIGITAL HPC	

Training Quality Principle: Attendance alone does not establish competency mastery. The department audits verifiable classroom implementation before certifying training completion.

9. Individual Professional-Development Action Plan Protocol

Every faculty member with an identified development priority receives an individualized, time-bound action plan:

Action Plan Component	Mandatory Documentation Specification
Educator Identity	Faculty name, institutional designation, and teaching allocation.
Identified Development Need	Specific pedagogical practice or technical skill requiring enhancement.
Baseline Diagnostic Evidence	Classroom observation notes, self-assessment ratings, or student work samples.

Measurable Growth Objective	Explicit, observable statement of intended instructional improvement.
Structured Action Plan	Targeted training, peer observation, mentor coaching, or self-paced study.
Institutional Support Provided	Allocated resources, software licenses, mentor pairing, or scheduling relief.
Implementation Timeline	Agreed inception date, midpoint milestone check, and final review date.
Observable Success Indicator	Concrete classroom evidence demonstrating enhanced instructional competence.
Verification Evidence	Revised lesson plans, observation rubrics, or improved student assessment data.
Review Outcome Verdict	Fully Achieved / Partially Achieved / Extended Support Required.

> **ACTION PLAN EXEMPLAR**
> *Exemplar Measurable Action Plan:*
> *• Development Need: Strengthen formative diagnostic checks during laboratory coding sessions.*
> *• Planned Action: Attend rubric-design workshop, embed terminal exit-check tickets in Class IX labs, and undergo peer observation.*
> *• Success Evidence: Documented diagnostic spreadsheets, rubric sheets with individualized student feedback, and verified elimination of recurring syntax errors on subsequent tests.*

10. Follow-Up and Evidence of Instructional Improvement

Professional development efficacy is assessed longitudinally through tangible changes in classroom delivery and student performance. Follow-up reviews verify:

- **Baseline vs. Post-Intervention:** Comparing pre-training observation notes with post-training instructional execution.
- **Applied Pedagogical Artifacts:** Reviewing updated lesson plans, newly introduced worksheets, and rubric designs.
- **Student Achievement Impact:** Examining whether targeted educator upskilling led to reduced student error rates and higher lab completion rates.
- **Sustained Practice:** Verifying that newly acquired teaching techniques become regular classroom routines rather than one-off demonstrations.

11. Departmental Competency Mapping and Strategic Review

The Head of Department aggregates individual evaluations to establish department-wide strategic priorities, addressing six foundational governance questions:

11.1 Curriculum Competency Match: Does every educator possess the required conceptual and technical mastery for their assigned grades?

11.2 Departmental Strengths vs. Gaps: Which instructional domains are securely established across the team, and which require collective upskilling?

11.3 Training Alignment: Do scheduled professional-development programs address genuine departmental instructional deficits?

11.4 Classroom Translation: Is institutional training actively translating into enhanced student machine engagement and higher output quality?

11.5 Resource & Mentoring Needs: What hardware upgrades, software licenses, or peer-mentoring structures are required to accelerate growth?

11.6 Longitudinal Accountability: What specific evidentiary benchmarks will substantiate departmental progress at the next audit cycle?

12. Departmental Monitoring Register

The department maintains a centralized live tracking register to audit educator progress across agreed development actions:

Educator Name	Review Date	Identified Key Strength	Priority Development Need	Agreed Action Plan	Target Timeline	Current Status
Mehraj U Din Mir	2026-04-15	Advanced Python & AI Architectures	Cloud deployment for student web apps	Develop containerized test lab workflow	Term-II (Oct 2026)	**In Progress**
Faculty Member A	2026-04-18	CBSE Board CS Syllabus Mastery	Granular rubric-based practical scoring	Peer observation & rubric calibration clinic	Mid-Term (Aug 2026)	**Completed**
Faculty Member B	2026-05-02	Interactive Block-Based Pedagogy	Introductory Python text syntax transition	Complete Python fundamentals upskilling	Term-I (Jul 2026)	**Completed**
Faculty Member C	2026-05-10	Digital Design & Creative Tools	Automated spreadsheet formula logic	Attend Excel advanced formulas workshop	Term-II (Nov 2026)	**In Progress**
Faculty Member D	2026-05-15	Hardware Maintenance & Networking	Classroom instructional pacing & timing	Structured lesson delivery mentoring with HOD	Term-I (Jun 2026)	**Evidence Pending**
Faculty Member E	2026-06-01	Primary Digital Literacy Instruction	Unplugged computational thinking games	Curate unplugged puzzle bank for Grade III-V	Term-II (Dec 2026)	**Not Started**

Status Categories: Not Started | In Progress | Evidence Pending | Completed | Further Support Required. Status reflects authenticated progress verified at each review cycle.

13. Evidence File Index for the Academic Audit

The department maintains a centralized, confidential archive comprising ten primary documentary streams:

13.1 Faculty Profiles & Verified Credentials: Official resumes, verified degree certificates, and subject allocation logs.

13.2 Professional Training Certifications: Accredited completion certificates from CBSE, IITs, and industry bodies.

13.3 Completed Faculty Self-Assessments: Annual self-assessment rubrics with documented evidence annotations.

13.4 Departmental Competency Matrices: Consolidated mapping charts identifying faculty strengths and growth needs.

13.5 Classroom Observation Dossiers: Formal observation rubrics, diagnostic notes, and agreed action plans.

13.6 Teaching Demonstrations & Sample Lessons: Lesson plans, demo courseware, and peer critique records.

13.7 Individual PD Action Plans: Targeted development plans detailing objectives, timelines, and resources.

13.8 Classroom Application Verification: Post-training lesson plans, worksheets, and student artifacts substantiating impact.

13.9 Internal Training & Mentoring Logs: Minutes and attendance logs from departmental workshops and coding clinics.

13.10 Longitudinal Progress Review Files: Annual review summaries certifying resolution of identified development needs.

Confidentiality & Compliance: Educator records are maintained securely, accessible strictly to authorized academic auditors and school leadership.

14. Expected Outcome and Departmental Commitment

This policy establishes a systematic, evidence-based approach to faculty capacity building across the ICT Department at Green Valley Educational Institute. It bridges professional development with measurable instructional improvement, purposeful technology integration, robust assessment literacy, and proactive responses to student learning needs.

Departmental Commitment: The ICT Department commits to utilizing competency mapping as a continuous improvement engine, systematically connecting professional learning with exemplary classroom practice and substantiated student achievement.

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